

Case Study: Deep Learning for Brain Tumour Detection Using MRI Scans

Title

Deep Learning-Based Brain Tumour Detection and Classification Using Convolutional Neural Networks (CNNs)

Introduction

Brain tumors are one of the most dangerous neurological diseases. Early diagnosis can significantly improve patient survival rates. Traditionally, radiologists manually examine MRI (Magnetic Resonance Imaging) scans to identify tumors. However, manual diagnosis is time-consuming and prone to human error.

Deep Learning, particularly **Convolutional Neural Networks (CNNs)**, can automatically detect and classify brain tumors with high accuracy by learning patterns from MRI images.

Problem Statement

Hospitals receive thousands of MRI scans every day. Manual examination by radiologists:

- Takes a long time.
- Requires expert knowledge.
- May result in incorrect diagnosis.
- Can delay treatment.

The goal is to build a Deep Learning model that automatically detects whether a brain tumor is present and identifies its type.

Objectives

- Detect brain tumors from MRI images.
- Classify tumors into different categories.
- Improve diagnostic accuracy.
- Reduce diagnosis time.
- Assist doctors in medical decision-making.

Dataset

Example Dataset:

Brain MRI Images Dataset

Contains approximately 7,000 MRI images.

Classes include:

- Glioma
- Meningioma
- Pituitary Tumor
- No Tumor

Image Size:

- 224×224 pixels

Image Format:

- JPG / PNG

Workflow:

MRI Images



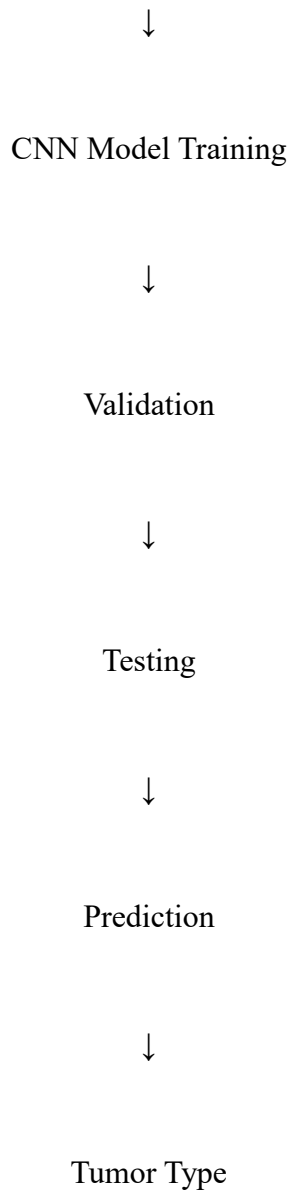
Data Collection



Image Preprocessing



Data Augmentation



Data Preprocessing

Before training:

- Resize all images to 224×224 .
- Normalize pixel values ($0-255 \rightarrow 0-1$).
- Remove corrupted images.
- Split data:
 - Training: 70%
 - Validation: 15%
 - Testing: 15%

Data Augmentation

To improve model performance:

- Rotation
- Horizontal Flip
- Vertical Flip
- Zoom
- Brightness Adjustment

This increases dataset diversity and reduces overfitting.

CNN Layers Explanation

Layer	Purpose
Convolution	Detects image features such as edges and shapes
ReLU	Introduces non-linearity
Max Pooling	Reduces image dimensions while keeping important features
Flatten	Converts feature maps into a 1D vector
Dense Layer	Learns relationships between extracted features
Softmax	Predicts the probability of each tumor class

Training Parameters

Parameter	Value
Epochs	30
Batch Size	32
Optimizer	Adam
Learning Rate	0.001
Loss Function	Categorical Cross-Entropy
Activation	ReLU, Softmax

Model Performance

Example Results:

Metric	Value
Training Accuracy	99%
Validation Accuracy	97%
Testing Accuracy	98%
Precision	97%
Recall	98%
F1 Score	97.5%

Confusion Matrix (Example)

Actual \ Predicted Glioma Meningioma Pituitary No Tumor

Glioma	120	3	2	1
Meningioma	4	118	2	1
Pituitary	2	1	122	0
No Tumor	1	0	2	125

Advantages of Using Deep Learning

- High accuracy in image classification.
- Detects subtle patterns that humans may miss.
- Reduces diagnosis time.
- Supports radiologists in clinical decision-making.

Challenges

- Requires a large amount of labeled MRI data.
- Training deep models demands significant computational resources (GPUs).
- Model performance may vary with image quality.
- Clinical deployment requires rigorous validation and regulatory approval.

Applications

- Hospital diagnostic support systems.
- Medical imaging centers.
- Telemedicine platforms.
- AI-assisted radiology.
- Medical research.

Technologies Used

- Python
- TensorFlow / Keras
- OpenCV
- NumPy
- Matplotlib
- Google Colab or Jupyter Notebook

Future Enhancements

- Use transfer learning models such as ResNet50, EfficientNet, or DenseNet for improved accuracy.
- Develop a web or mobile application for real-time predictions.
- Integrate explainable AI techniques (e.g., Grad-CAM) to highlight image regions influencing predictions.

Conclusion

This case study demonstrates how **Deep Learning**, particularly **Convolutional Neural Networks (CNNs)**, can significantly improve brain tumor detection from MRI scans. By automatically extracting features from medical images, CNNs provide highly accurate and efficient classification, helping healthcare professionals make faster and more reliable diagnoses. Although challenges such as data availability and computational requirements remain, deep learning has become a valuable tool in modern medical imaging and continues to enhance patient care through AI-assisted diagnosis.